Personal Page:
danilopezzi.github.io

Danilo Pezzi

Postdoctoral Researcher

Mathematician specialized in numerical optimization and machine learning, with expertise in data-driven and explainable models for imaging. I develop algorithms that combine classical optimization methods with modern machine learning techniques. I leverage my scientific background to design robust and interpretable solutions. I am seeking opportunities outside academia to apply these skills to real-world problems and impactful applications.

Work Experience

Postdoctoral Research Fellow – University of Modena and Reggio Emilia. December 2023 - Present

- Research in the field of numerical optimization, inverse problems in imaging (denoising, deblurring, super-resolution) and machine learning. Design and implementation of the experimental part for scientific publications, with responsibility for writing and presenting the results.
- Contributed to an industrial project on autonomous driving, eventually taking technical leadership of the thermal imaging workstream focused on object detection and semantic segmentation. Evaluation, benchmarking, and optimization of state-of-the-art algorithms. Development of generative AI techniques for data synthesis and training data augmentation.
- Presentation of research work at international conferences and national events. Organization of two special sessions at SIMAI 2025.
- Served as a reviewer for international scientific journals: evaluation of scientific content, experimental validity, and clarity of presentation.
- Scientific outreach with interactive presentations on machine learning and artificial intelligence at public engagement events.
- Over 200 hours of mathematics teaching experience for students at various levels.

Education

Ph.D. in Mathematics – University of Parma. November 2020 - March 2024

- Thesis: Learning Variational Models via Bilevel Optimization and Unfolded Algorithms.
- Three-month research stay at the i3S laboratory (Université Côte d'Azur, France). Collaboration on imaging problems for microscopy.

Master's Degree in Mathematics – University of Modena and Reggio Emilia October 2018 - July 2020

- Thesis: Convex Analysis and Optimization: Splitting Methods.
- Final grade: 110/110 cum laude.

Bachelor's Degree in Mathematics – University of Modena and Reggio Emilia October 2015 - July 2018

- Thesis: Reed-Solomon Codes in Compact Discs.
- Final grade: 110/110 cum laude.

Skills

Programming languages: MATLAB, Python (Pytorch, Numpy, Scipy).
Tools: LaTeX, Git (familiarity).
Research topics: numerical optimization, inverse problems, imaging, machine learning, deep learning.